

BrainHIVE Retrieval Reproduction

Corrected reproduction of the B32 versus B32+VAE ablation on THINGS-EEG

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Paper	Learning Brain Representation with Hierarchical Visual Embeddings (ICLR 2026)
Target	Intra-subject 200-way zero-shot retrieval, B32 versus B32+VAE
Repository	github.com/HUI-HUI-lab/brainhive-retrieval-reproduction
Corrected run	16 July 2026 - 10 subjects, 20 formal runs

Outcome: canonical B32 features raised the reproduced mean to 48.8%/81.0% Top-1/Top-5. B32+VAE reached 69.7%/92.95%. The reproduced VAE gain (+20.9/+11.95 points) closely matches the paper's detailed Table 7 gain (+22.5/+11.1 points).

Scope: this is one representative ablation. Reconstruction, diffusion-prior training, inter-subject retrieval, and other visual encoders are outside scope.

1. Paper and reproduction target

BrainHIVE aligns brain signals with a fused hierarchy of visual representations. The selected experiment asks whether adding low-level SDXL VAE information to semantic LAION CLIP ViT-B/32 features improves 200-way THINGS-EEG retrieval. The paper reports aggregate values in Table 3 and detailed per-subject values in Table 7. Table 7 is used for the primary corrected comparison because it exposes the same ten paired subjects.

2. Original and reproduced settings

Component	Paper	Corrected reproduction
Task	THINGS-EEG intra-subject, 200-way	Same; all ten subjects
EEG	17 O+P channels; trials averaged	Same; validated 17x250 tensors
Targets	LAION B32; B32+SDXL VAE	Canonical B32 512-D; verified VAE 1024-D
Training	25 epochs; AdamW; LR 5e-4; batch 1024	Same; one seed (2025); 10-step warmup
Metric	Top-1/Top-5 over 200 candidates	Same; evaluation batch 200
Compute	Single RTX 5090 reported for contrastive stage	CPU FP32; four concurrent workers
Runs	Averages; public launcher repeats 3x	One run/subject/setting

Each subject contained 16,540 averaged training examples and 200 averaged test examples. Preflight checks verified sample counts, dimensions, finite values, unique image IDs, and EEG-to-embedding alignment before training.

Controlled modifications

The original LLM is irrelevant to this retrieval stage. One seed replaced repeated runs. CPU FP32 was used because both shared GPUs were occupied; optimizer, batch size, epochs, channels, warmup, loss, and evaluation remained fixed.

3. Cache audit and failed first attempt

The first ten-subject run used the public Hugging Face files named as B32 and produced 21.6%/53.05%. B32+VAE produced 50.2%/82.55%. Rather than attributing the entire gap to random seed or precision, the visual targets were inspected directly.

The public B32 cache was severely collapsed: unrelated image vectors were almost parallel. It should not be used for this experiment without validation.

Diagnostic	Canonical / trusted B32	Public HF B32
Random-pair cosine median	0.3264	0.9943
Random-pair cosine mean	0.3300	0.9915
Normalized mean-vector norm	0.5745	0.9957
Same image vs fresh official extraction	0.99992	0.20091
Nearest-neighbor row recovery (sample 256)	Expected semantic structure	0/256 correct rows

Image IDs, ordering, dimensions, and row counts were correct, so a simple shard-order bug was ruled out. A fresh embedding for the same source image matched the trusted cache almost exactly but not the public cache. The exact upstream generation or upload error cannot be recovered from the available metadata.

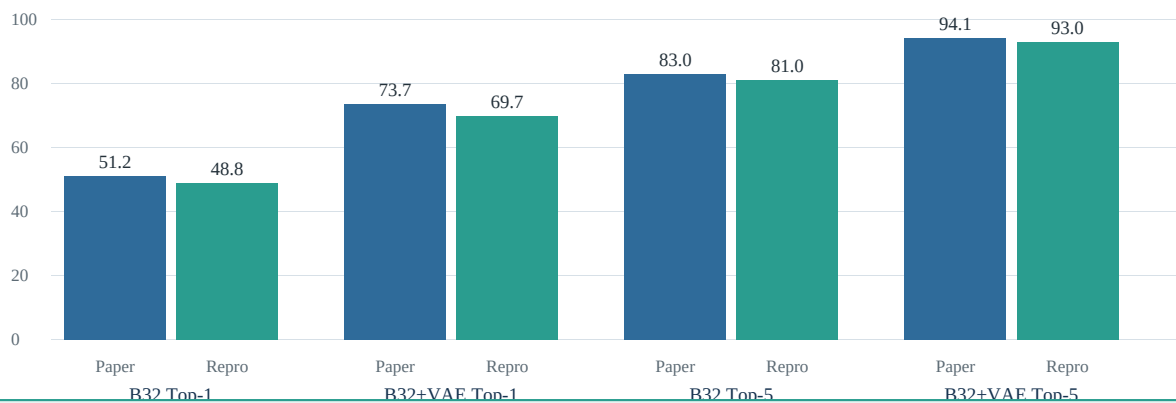
Corrective action

The corrected run used a trusted read-only B32 cache verified against the canonical model and canonical preprocessing. The VAE cache was retained because its random-pair cosine median (0.0426) and alignment checks were healthy. The repository now includes a mandatory collapse detector and excludes public B32 from default downloads.

4. Corrected results

Source / setting	Top-1 (%)	Top-5 (%)	VAE gain (pp)
Paper Table 7 - B32	51.2	83.0	-
Paper Table 7 - B32+VAE	73.7	94.1	+22.5 / +11.1
Corrected reproduction - B32	48.8	81.0	-
Corrected reproduction - B32+VAE	69.7	92.95	+20.9 / +11.95

Ten-subject retrieval accuracy (%)



The corrected absolute gap is small: **-2.4/-2.0 points for B32** and **-4.0/-1.15 points for B32+VAE** relative to Table 7. Exact agreement is not required by the assignment, and the paired improvement is the main target.

5. Per-subject result and interpretation

Subj	B32 T1	B32 T5	B32+VAE T1	B32+VAE T5	T1 gain	T5 gain
1	38.5	73.0	62.0	89.0	+23.5	+16.0
2	46.0	76.5	70.5	93.0	+24.5	+16.5
3	48.5	80.5	68.5	92.5	+20.0	+12.0
4	51.5	83.5	58.5	89.0	+7.0	+5.5
5	38.5	71.0	60.5	89.0	+22.0	+18.0
6	52.0	85.0	72.5	94.0	+20.5	+9.0
7	47.0	80.0	68.5	92.5	+21.5	+12.5
8	61.5	88.0	81.0	97.0	+19.5	+9.0
9	41.0	79.0	72.0	95.0	+31.0	+16.0
10	63.5	93.5	83.0	98.5	+19.5	+5.0
Mean	48.8	81.0	69.7	93.0	+20.9	+12.0

Is the paper's conclusion supported?

Yes. B32+VAE improves both Top-1 and Top-5 for every reproduced subject. The mean Top-1 gain differs from the paper by only 1.6 points, and the mean Top-5 gain differs by 0.85 points. This is a clear reproduction of the reported performance trend and approximately supports the paper's conclusion that low-level VAE features complement high-level CLIP semantics.

Remaining differences

The remaining 1-4 point absolute gaps are consistent with one seed instead of repeated runs, CPU FP32 numerical behavior, dependency and code-snapshot drift, and small internal differences between the paper's aggregate tables. The corrected result is much closer to the paper than to the failed public-cache run, confirming that target-feature quality was the dominant discrepancy.

6. AI-assisted workflow

Codex assisted with repository inspection, experiment design, code changes, SSH diagnostics, cache forensics, data validation, job monitoring, aggregation, documentation, and PDF generation. Human decisions selected the target, approved server access, authorized execution, and accepted the CPU fallback when shared GPUs were busy.

Purpose	Representative prompt / action
Planning	Inspect BrainHIVE and design a minimal Table 3 retrieval reproduction.
Execution	Check resources, validate data, run one subject, then run all ten subjects safely.
Diagnosis	Why is the reproduction far below the paper? Compare embeddings before changing training.
Cache audit	Recompute one canonical B32 feature and test whether the public cache is permuted or collapsed.
Correction	Use verified B32, rerun B32 and B32+VAE for all subjects, aggregate, and update deliverables.

7. Runtime, cost, and reproducibility

Resource	Recorded / estimated value
Corrected formal runs	20/20 successful; 25 epochs each
Matrix wall time	Approximately 17 min 47 s with four CPU workers
Summed runtime	3,867 s; mean 193.35 s/run
AI-assisted time	Approximately 4 hours across planning, debugging, execution, and writing
AI token usage	Approximately 200k tokens; account export unavailable
Incremental AI cost	Not separately metered; Codex account/subscription usage

Commands

```
python reproduction/scripts/validate_embeddings.py --data-root DATA
python reproduction/scripts/validate_data.py --data-root DATA --subjects 1 --models
CLIP-ViT-B-32-laion2B-s34B-b79K vae
PRECISION=fp32 USE_CPU=true MAX_PARALLEL=4 bash reproduction/scripts/run_matrix.sh DATA RESULTS
1,2,3,4,5,6,7,8,9,10 2025
```

8. Limitations and deliverables

Only one seed was run. The exact internal paper snapshot and original cached features were unavailable. The corrected run reused a trusted canonical cache after direct official-model verification rather than regenerating all 16,740 images. CPU FP32 differs from the paper's GPU execution. Approximate AI token, time, and cost values should be replaced with account-exported figures if available.

The repository contains runnable code, pinned dependencies, safe data-preparation instructions, cache and alignment validators, CPU/GPU launch commands, aggregation scripts, machine-readable results, this PDF, HowTo.md, and a reusable research-server skill. Large datasets, caches, checkpoints, credentials, private keys, and private infrastructure details are not committed.

Repository

<https://github.com/HUI-HUI-lab/brainhive-retrieval-reproduction>

References

- [1] Zheng, J. et al. *Learning Brain Representation with Hierarchical Visual Embeddings*. ICLR 2026. [arXiv:2602.07495](https://arxiv.org/abs/2602.07495)
- [2] BrainHIVE source. github.com/ssshamiii/Brain-HIVE
- [3] THINGS-EEG. [Hugging Face dataset](#)
- [4] Public visual cache inspected in the failed first attempt. [Hugging Face dataset](#)